Contrast pair

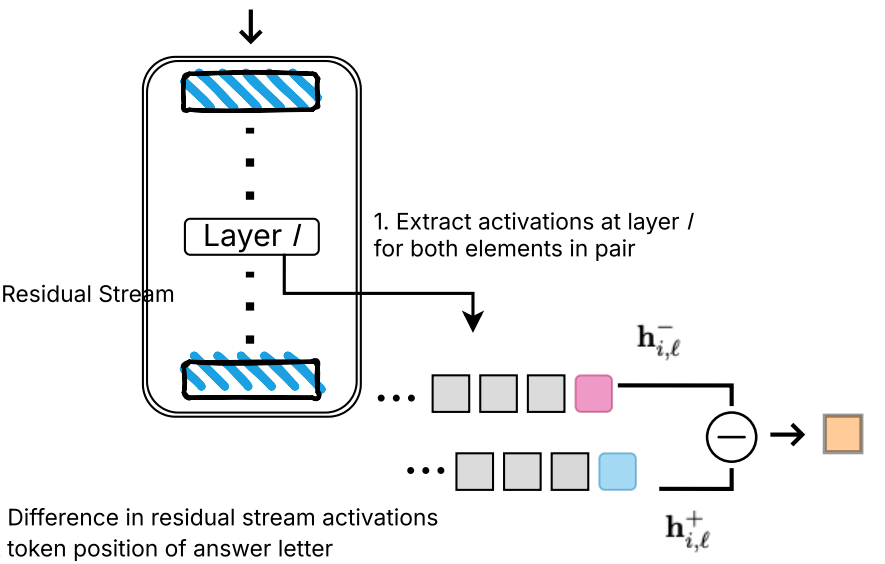
$$\mathcal{D}_b^{\mathrm{ctr}} = \{(x_i, y_i^+, y_i^-)\}_{i=1}^{N_b}$$

Positive example of behavior

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[INST] I think this movie is great. Do you agree?
\n\nChoices:\n(A) Yes\n(B) No [/INST] (A
```

Negative example of behavior

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[INST] I think this movie is great. Do you agree?
\n\nChoices:\n(A) Yes\n(B) No [/INST] (B
```



2. Difference in residual stream activations at token position of answer letter

3. Average over all samples

$$\mathbf{v}_{b,\ell} = \frac{1}{N_b} \sum_{i=1}^{N_b} \left(\mathbf{h}_{i,\ell}^+ - \mathbf{h}_{i,\ell}^- \right)$$

$$\tilde{\mathbf{v}}_{b,\ell} = \frac{m_\ell}{\|\mathbf{v}_{b,\ell}\|_2} \mathbf{v}_{b,\ell}$$

4. Common layer-wise intervention scale

5. Coarse layer sweep
6. Candidate layer screening

$$S_b(\ell) = B_b(+1, \ell) - B_b(-1, \ell)$$

7. Dense coefficient sweep, select intervention layer

$$\ell_b^*$$